

L13: Topology Inference Using Network Tomography

Let us now focus on topology inference using *network tomography*. In medical imaging, "tomography" refers to methods that observe the internal structure of a system (e.g., brain tissue) using only measurements from the exterior or "periphery" of the system. In the context of networks, the internal structure refers to the topology of the network (both nodes and edges), while the "periphery" is few nodes that are observable and that we can utilize to make measurements.

For instance, in the following tree network the observable nodes are the root (blue) and the leaves (yellow) -- all internal nodes (green) and the edges that interconnect all nodes are not known.

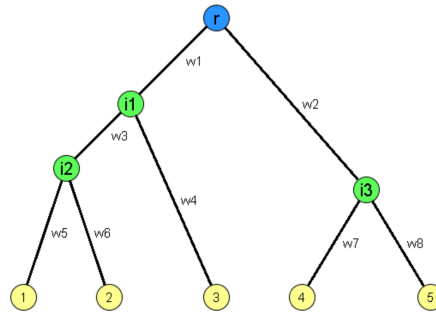


Image source: Kolaczyk, Eric D. Statistical Analysis of Network Data: Methods and Models (2009) Springer Science+Business Media LLC.

To simplify, we will describe network tomography in the context of *computer networks*, where the nodes represent computers or routers and the edges represent transmission links. Please keep in mind however that network tomography methods are quite general and they are also applicable in other contexts, such as the *inference of phylogenetic trees in biology*.

Let us first review a basic fact about computer networks. When a packet of size L bits is transmitted by a router (or computer) on a link of capacity C bits-per-second, the transmission takes L/C seconds. So, if we send two packets of size L at that link, the second packet will have to wait at a router buffer for the transmission of the first packet -- and that waiting time is L/C .

We can use this fact to design a smart measurement technique called "*sandwich probing*". Consider the previous tree network. Suppose that the root node sends three packets $P1, P2, P3$ at the same time. Packets $P1$ and $P3$ are small (the minimum possible size) and they are destined to one of the observable nodes, say R_i , while the intermediate packet $P2$ is large (the largest possible size) and it is destined to another observable node R_j . The packets are timestamped upon transmission, and the receiving node R_i measures the end-to-end transfer delay that the two small packets experienced in the network.

If the destinations R_i and R_j are reachable from the root through completely different paths (e.g., such as the leaves 1 and 5 in the previous tree), packet $P3$ will never be delayed due to packet $P2$ because the latter follows a different path. So, the transfer delays of the two small packets $P1$ and $P3$ will be very similar. The absolute difference of those two transfer delays will be close to 0.

On the contrary, if R_i and R_j are reachable through highly overlapping paths (such as the leaves 1 and 2 in the previous tree), packet $P3$ will be delayed by the transmission delay of packet $P2$ at every intermediate router in the overlapping segment of the two paths. The more the intermediate routers in the common portion of the paths to R_i and R_j , the larger the extra transfer delay of packet $P3$ relative to the transfer delay of packet $P1$.

Let us denote by $d_{i,j}$ the difference between the transfer delays of packets $P1$ and $P3$ when they are sent to destination R_i , while the large packet $P2$ is sent to destination R_j . This metric carries some information about the overlap of the network paths from the root node to R_i and R_j . In practice, we would not send just one "packet sandwich" for each pair of destinations -- we would repeat this 1000s of times and measure the average value of $d_{i,j}$.

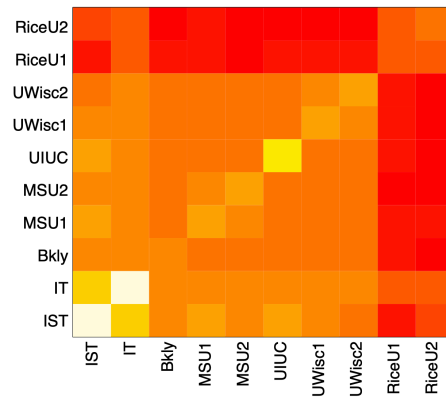


Image source: Kolaczyk, Eric D. *Statistical Analysis of Network Data: Methods and Models* (2009) Springer Science+Business Media LLC.

The previous figure visualizes these average delay differences for an experiment in which about 10,000 "packet sandwiches" were sent from a computer at Rice University in Texas to ten different computers: two of them also at Rice, others at other US universities and two (IST and IT) in Portugal. The darker color represents lower values (closer to 0), while the brighter color represents higher values. Note, for instance, that when we send the small packets to one of the Rice destinations and the large packet to a destination outside of Rice, the delay difference is quite low (relative to the case that both destinations are outside of Rice).

Now that we understand that we can use $d_{i,j}$ as a metric of "path similarity" for every pair of destinations, we can apply a *hierarchical clustering algorithm* to infer a *binary tree*, rooted at the source of the packet sandwiches, while all destinations reside at the leaves of the tree. Recall that we used hierarchical clustering algorithms in Lesson-7 for community detection. Here, our goal is to identify the binary tree that "*best explains*" the delay differences $d_{i,j}$, across all possible pairs i and j .

The algorithm proceeds iteratively, creating a new internal tree node in each iteration. At the first iteration, it identifies the two leaves i and j that have the *largest* delay difference (i.e., the two destinations that appear to have the highest network path overlap), and it groups them together by creating an *internal tree node* $a(i, j)$ that becomes the parent of nodes i and j . Then, the delay difference between the new node $a(i, j)$ and any other leaf k is calculated as the average of the delay differences $d_{i,k}$ and $d_{j,k}$ (i.e., we use "average linking"). The two leaves i and j are marked as "covered", so that they are not selected again in subsequent iterations.

The algorithm proceeds until all nodes, both the original leaves and the created intermediate nodes, are "covered".

In the following figure we show what happens when we apply this method in the previous delay difference data. The top visualization shows the actual network paths from the source node at Rice to the ten destinations. The bottom visualization shows the result of the hierarchical clustering algorithm we described earlier.

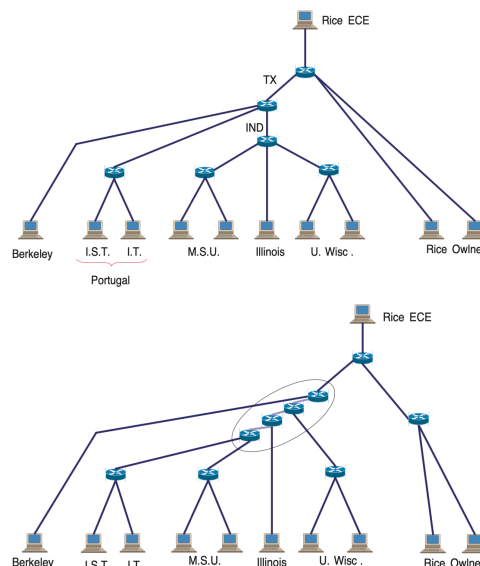


Image source: Kolaczyk, Eric D. *Statistical Analysis of Network Data: Methods and Models* (2009) Springer Science+Business Media LLC.

Note that a first difference between the "ground truth" network and the inferred network is that the latter is a binary tree, while the former includes branching nodes (routers) with more than two children. There is also a router (IND) that is completely missing from the inferred network, probably because it is so far that it does not cause a measurable increase in the delay of packet P_3 .

Food For Thought

- a) The metric we have introduced here is based on delay variations using "packet sandwiches". Can you think of other ways to probe a network in order to measure the topological overlap of different paths?
- b) The network inference method we used here is based on hierarchical clustering. Can you think of other network inference methods we could use instead? (Hint: remember what we did in Lesson-12 with the dendrogram of the HRG model)